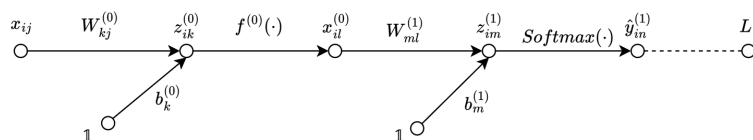


Neural_Networks

Task	Convolutional	Transformer
Dense Prediction (segmentation & depth estimation)	FCN, U-Net, monoDepth	DPT, SETR
Image Recognition	ResNet-50, EfficientNet	ViT, DinoV2
Object Detection	YOLO, Faster R-CNN, CenterNet, FCOS	DETR
Image Captioning	CNN+LSTM,	CLIP
Keypoint detection (local feature matching)	Superpoint, superglue	Loftr

	Low dimensional states	High-dimensional states
Static dataset	Classic supervised learning (ML)	Deep supervised learning (DL)
Agent/environment interaction	Tabular reinforcement learning	Deep reinforcement learning

MLP for Classification



1. A simple MLP with a single hidden layer, i.e. one-hidden-layer neural network, with cross-entropy loss. In this model the output layer is a linear layer which will be explained later.
2. Note: the i index is specially reserved for the sample index, and that the indices j, k, l, m and n for the layer widths are all just dummy variables (no special meaning whatsoever), the 'input' is **not** a layer although historically it was referred to as the 'input layer'.

Forward variables

Symbol	meaning	Shape	
N	batch size	-	
x	input	(N, C^{input})	(batch_size, input_size)
$z^{(0)}$	hidden affine layer	$(N, C^{(0)})$	(batch_size, hidden_size)
$x^{(0)}$	hidden layer	$(N, C^{(0)})$	(batch_size, hidden_size)
$z^{(1)}$	output affine layer	$(N, C^{(1)})$	(batch_size, output_size)
$y^{(1)}$	output layer	$(N, C^{(1)})$	(batch_size, output_size)

Model weights

Symbol	meaning	Shape	
$W^{(0)}$	hidden layer weights	$(C^{(0)}, C^{(input)})$	(hidden_size, input_size)
$b^{(0)}$	hidden layer bias	$(C^{(0)}, 1)$	(hidden_size, 1)
$W^{(1)}$	output layer weights	$(C^{(1)}, C^{(0)})$	(output_size, hidden_size)
$b^{(1)}$	output layer bias	$(C^{(1)}, 1)$	(output_size, 1)

3. Note: input_size refers to the number of input features. It is different from the number of data samples, which in this case correspond to the batch_size because we are using mini-batch gradient descent for training. For stochastic gradient descent

the number of data samples used in each iteration to update the weights would be 1.

4. Scalar form

1. Forward

$$1. z_{ik}^{(0)} = \sum_j W_{kj}^{(0)} x_{ij} + b_k^{(0)}$$

$$2. x_{il}^{(0)} = f^{(1)}(z_{ik}^{(0)})$$

$$3. z_{im}^{(1)} = \sum_l W_{ml}^{(1)} x_{il}^{(0)} + b_m^{(1)}$$

$$4. y_{in}^{(1)} = \text{Softmax}(z_{im}^{(1)}) = \frac{e^{z_{in}^{(1)}}}{\sum_m e^{z_{im}^{(1)}}}$$

$$5. L = \frac{1}{N} \sum_i \sum_n y_{in} \log \frac{y_{in}}{y_{in}^{(1)}}$$

6. where z is also called the logits, the output of the model before being normalized into probabilities.

7. The loss is a distance measure of how far the prediction score is from the ground truth score. To compare distances of two probability distributions, the Kullback-Leibler divergence is used.

2. Backward

1. To distinguish between predicted and ground truth probabilities y 's let the prediction, $y_{in}^{(1)} = \hat{y}_{in}$. Then the classification loss is

$$2. L = \frac{1}{N} \sum_i \sum_n (y_{in} \log y_{in} - y_{in} \log \hat{y}_{in})$$

3. For one-hot encoding of the ground truth probabilities the loss function will be reduced to cross-entropy loss,

$$4. L = \frac{1}{N} \sum_i \sum_n -y_{in} \log \hat{y}_{in}$$

$$5. \frac{\partial L}{\partial z_{im}^{(1)}} = \frac{1}{N} \sum_n \frac{\partial L}{\partial y_{in}} \frac{\partial y_{in}}{\partial z_{im}^{(1)}} = \frac{1}{N} (\hat{y}_{im} - y_{im})$$

3. Remark on derivative of CE Loss to the logits: To obtain $\frac{\partial L}{\partial z_{im}^{(1)}}$ the Jacobian for the Softmax vector activation is given by,

$$4. \frac{\partial \hat{y}_{in}}{\partial z_{im}^{(1)}} = \frac{\partial}{\partial z_{im}^{(1)}} \left(\frac{e^{z_{in}^{(1)}}}{\sum_m e^{z_{im}^{(1)}}} \right) = \begin{cases} \hat{y}_{im}(1 - \hat{y}_{im}), & \text{if } n = m \\ -\hat{y}_{im}\hat{y}_{in}, & \text{if } n \neq m \end{cases}$$

$$5. \frac{\partial L}{\partial y_{in}} = -\frac{y_{in}}{\hat{y}_{in}}$$

6. Combining the above two equations,

$$\begin{aligned} \frac{\partial L}{\partial z_{im}^{(1)}} &= \frac{1}{N} \left(\sum_{n \neq m} -\frac{y_{in}}{\hat{y}_{in}} (-\hat{y}_{im}\hat{y}_{in}) + -\frac{y_{im}}{\hat{y}_{im}} \hat{y}_{im}(1 - \hat{y}_{im}) \right) \\ &= \frac{1}{N} \left(\sum_{n \neq m} y_{in}\hat{y}_{im} - y_{im} + y_{im}\hat{y}_{im} \right) \\ 7. &= \frac{1}{N} \left(\hat{y}_{im} \underbrace{\sum_n y_{in}}_{=1 \Rightarrow \text{one-hot}} - y_{im} \right) \\ &= \frac{1}{N} (\hat{y}_{im} - y_{im}) \end{aligned}$$

5. Matrix form

1. Forward

$$1. z^{(0)} = x \cdot W^{(0)T} + b^{(0)}$$

$$2. x^{(0)} = f^{(0)}(z^{(0)})$$

$$3. z^{(1)} = x^{(0)} \cdot W^{(1)T} + b^{(1)}$$

$$4. y_j^{(1)} = \text{Softmax}(z_j^{(1)}) = \frac{e^{z_j^{(1)}}}{\sum_j e^{z_j^{(1)}}}$$

$$5. L = \frac{\iota_N^T (-y \odot \log \hat{y}^{(1)}) \cdot \iota_C}{N}$$

2. Backward

$$1. \frac{\partial L}{\partial z^{(1)}} = \frac{\hat{y}^{(1)} - y}{N}$$

$$2. \frac{\partial L}{\partial x^{(0)}} = \frac{\partial L}{\partial z^{(1)}} \cdot W^{(1)}$$

$$3. \frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial z^{(1)}}^T x^{(0)}$$

$$4. \frac{\partial L}{\partial b^{(1)}} = \frac{\partial L}{\partial z^{(1)}}^T \iota_N$$

$$5. \frac{\partial L}{\partial z^{(0)}} = \frac{\partial L}{\partial x^{(0)}} \odot f'(z^{(0)})$$

Activation functions

1. Code: mytorch/activations.py

```

import numpy as np

class Activation:
    def __init__(self):
        self.x = None
        self.update_weights = False

    def forward(self, z):
        raise NotImplementedError()

    def backward(self, dLdx):
        raise NotImplementedError()

class Identity(Activation):
    def forward(self, z):
        self.x = z
        return self.x

    def backward(self, dLdx):
        dx dz = np.ones_like(self.x)
        dLdz = dLdx * dx dz
        return dLdz

class ReLU(Activation):
    def forward(self, z):
        self.x = np.maximum(z, np.zeros_like(h))
        return self.x

    def backward(self, dLdx):
        dx dz = np.zeros_like(self.x)
        dx dz[np.where(self.x > 0)] = 1
        dLdz = dLdx * dx dz
        return dLdz

class Tanh():
    """
    Modified for RNN Classifier.
    """
    def forward(self, z):
        # self.x = (np.exp(z) - np.exp(-z)) / (np.exp(z) + np.exp(-z))
        self.x = np.tanh(z) # more stable to overflow
        return self.x

    def backward(self, dLdx, h_t=None):
        if h_t is None:
            dx dz = 1 - self.x ** 2
        else:
            dx dz = 1 - h_t ** 2
        dLdz = dLdx * dx dz
        return dLdz

class Sigmoid(Activation):
    def forward(self, z):
        self.x = 1 / (1 + np.exp(-z))
        return self.x

    def backward(self, dLdx):
        dLdz = dLdx * self.x * (1 - self.x)
        return dLdz

class Softmax(Activation):
    def forward(self, z):
        exp_z = np.exp(z - np.max(z, axis = 1, keepdims=True))
        sum_exp_z = np.sum(exp_z, axis = 1, keepdims=True)
        self.x = exp_z / sum_exp_z
        return self.x

    def backward(self, dLdx):
        N = self.x.shape[0]
        C = self.x.shape[1]
        dLdz = np.zeros(shape=(N, C))
        for k in range(N):
            J = np.zeros(shape=(C, C))
            for i in range(C):
                for j in range(C):
                    if i == j:
                        J[i, j] = self.x[k, i] * (1 - self.x[k, i])
                    else:
                        J[i, j] = - self.x[k, i] * self.x[k, j]
            dLdz[k, :] = dLdx[k, :] @ J
        return dLdz

if __name__ == "__main__":
    z = np.random.randint(-5, 5, (3, 4))
    relu = ReLU()
    x = relu.forward(z)
    dLdx = np.random.randint(0, 5, size=x.shape)
    dLdz = relu.backward(dLdx)
    print("z = ", z)
    print("x = ", x)

```

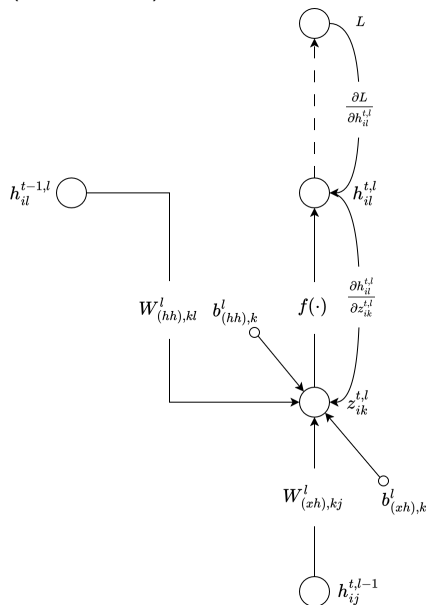
```

print("dLdx = ", dLdx)
print("dLdz = ", dLdz)
print("")
softmax = Softmax()
x = softmax.forward(z)
dLdz = softmax.backward(dLdx)
print("z = ", z)
print("x = ", x)
print("dLdx = ", dLdx)
print("dLdz = ", dLdz)

```

RNN

1. RNN Cell (CMU 11-785)



1.

2. An RNN Cell forward

1. takes as input: x or h_{prev_l} , h_{prev_t}
2. produces: h_{next} and cache: $(h_{\text{next}}, h_{\text{prev}_l}, h_{\text{prev}_t}, dh_{\text{next}})$ (for backprop, in most implementations)

3. An RNN Cell backward

1. takes as input, upstream gradient dh_{next} , h_{next} , h_{prev_l} , h_{prev_t}
2. produces: dx or dh_{prev_l} , dh_{prev_t}

4. Code: mytorch/rnn_cell.py

```

import numpy as np
from .activation import *

class RNNCell:

    def __init__(self, input_size, hidden_size):

        self.input_size = input_size
        self.hidden_size = hidden_size

        # Activation function for
        self.activation = Tanh()

        # hidden dimension and input dimension
        h = self.hidden_size
        d = self.input_size

        # Weights and biases
        self.W_ih = np.random.randn(h, d)
        self.W_hh = np.random.randn(h, h)
        self.b_ih = np.random.randn(h)
        self.b_hh = np.random.randn(h)

        self.dW_ih = np.zeros((h, d))
        self.dW_hh = np.zeros((h, h))
        self.db_ih = np.zeros(h)
        self.db_hh = np.zeros(h)

    def init_weights(self, W_ih, W_hh, b_ih, b_hh):
        self.W_ih = W_ih
        self.W_hh = W_hh
        self.b_ih = b_ih
        self.b_hh = b_hh

    def zero_grad(self):
        h, d = self.hidden_size, self.input_size

```

```

self.dW_ih = np.zeros((h, d))
self.dW_hh = np.zeros((h, h))
self.db_ih = np.zeros(h)
self.db_hh = np.zeros(h)

def __call__(self, x, h_prev_t):
    return self.forward(x, h_prev_t)

def forward(self, x, h_prev_t):
    h_next = h_prev_t @ self.W_hh.T + self.b_ih + x @ self.W_ih.T + self.b_hh
    out = self.activation.forward(h_next)
    return out

def backward(self, dh_next, h_next, h_prev_l, h_prev_t):
    dz = self.activation.backward(dh_next, h_next)
    batch_size = dh_next.shape[0]
    # 1) Compute the averaged gradients of the weights and biases
    self.dW_ih += dz.T @ h_prev_l / batch_size
    self.dW_hh += dz.T @ h_prev_t / batch_size
    self.db_ih += np.sum(dz.T, axis=1) / batch_size
    self.db_hh += np.sum(dz.T, axis=1) / batch_size

    # 2) Compute dx, dh_prev_t
    dx = dz @ self.W_ih
    dh_prev_t = dz @ self.W_hh

    return dx, dh_prev_t

```

2. RNN Classifier (CMU 11-785)

1. Example: many-to-one, multiple layers RNN Classifier
2. RNN Classifier forward,
3. RNN Classifier backward,
 1. An important principle for BPTT: very simple rule: for each node with multiple descendants in the computational graph, simply add up the delta vectors coming from all of the descendants.
4. Code: models/rnn_classifier.py

```

import numpy as np

from HW3.P1.mytorch.rnn_cell import *
from HW3.P1.mytorch.linear import *

class RNNPhonemeClassifier(object):
    """RNN Phoneme Classifier class."""

    def __init__(self, input_size, hidden_size, output_size, num_layers=2):
        self.input_size = input_size
        self.hidden_size = hidden_size
        self.num_layers = num_layers

        # TODO: Understand then uncomment this code :)
        self.rnn = [
            (
                RNNCell(input_size, hidden_size)
                if i == 0
                else RNNCell(hidden_size, hidden_size)
            )
            for i in range(num_layers)
        ]
        self.output_layer = Linear(hidden_size, output_size)

        # store hidden states at each time step, [(seq_len+1) * (num_layers, batch_size, hidden_size)]
        self.hiddens = []

    def init_weights(self, rnn_weights, linear_weights):
        """Initialize weights.

        Parameters
        -----
        rnn_weights:
            [
                [W_ih_l0, W_hh_l0, b_ih_l0, b_hh_l0],
                [W_ih_l1, W_hh_l1, b_ih_l1, b_hh_l1],
                ...
            ]

        linear_weights:
            [W, b]

        """
        for i, rnn_cell in enumerate(self.rnn):
            rnn_cell.init_weights(*rnn_weights[i])
        self.output_layer.W = linear_weights[0]
        self.output_layer.b = linear_weights[1].reshape(-1, 1)

    def __call__(self, x, h_0=None):
        return self.forward(x, h_0)

```

```

def forward(self, x, h_0=None):
    """RNN forward, multiple layers, multiple time steps.
    x: input sequence: (N, T, D)
    self.x: cache (for backprop) of input sequence: (N, T, D)
    h0: initial hidden state (optional), zeros if not given: (L, N, H)
    self.hiddens: cache (for backprop) of computed hidden states (h_next): (T+1, L, N, H)
    hiddens: computed hidden states for all the layers (L, N, H)
    h_next: hidden state (N, H)
    logits: output (N, H_out)
    """
    N, T, D = x.shape
    L = self.num_layers
    H = self.hidden_size

    if h_0 is None:
        hiddens = np.zeros((L, N, H), dtype=np.float64)
    else:
        L, _, H = h_0.shape
        assert L == self.num_layers, H == self.hidden_size
        hiddens = h_0
    self.hiddens.append(hiddens.copy())
    self.x = x
    for t in range(T):
        for l in range(L):
            # Compute h_prev_l
            if l == 0:
                h_prev_l = x[:, t, :]
            else:
                h_prev_l = hiddens[l - 1, :, :]
            # Compute h_prev_t
            h_prev_t = hiddens[l]
            # Perform forward on RNN Cell
            h_next = self.rnn[l].forward(h_prev_l, h_prev_t)
            # Update hiddens for next time step
            hiddens[l] = h_next

        self.hiddens.append(hiddens.copy())

    logits = self.output_layer.forward(h_next)
    return logits

def backward(self, delta):
    """RNN Back Propagation Through Time (BPTT).
    N: number of samples or batch size
    L: number of layers
    H: hidden dimension
    D: input dimension
    H_out: output dimension
    delta: Upstream gradient, dL/dy from linear layer before loss function
    i.e. in last RNN Cell, h_next -> y -> L: (N, H_out)
    self.hiddens: Contains h_next computed in each RNN Cell: (T+1, L, N, H)
    self.x: input sequence: (N, T, D)
    dh_next: contains upstream gradients: (L, N, H)
    for backprop, cache = (h_next, h_prev_l, h_prev_t, dh_next)
    """
    T = self.x.shape[1]
    L = self.num_layers
    H = self.hidden_size
    N, _ = delta.shape
    dh_next = np.zeros((L, N, H), dtype=np.float64)
    # Compute upstream gradient, dL/dh_next, for last RNN Cell
    dh_next[-1] = self.output_layer.backward(delta)
    # self.output_layer.backward(delta)
    for t in reversed(range(T)):
        for l in reversed(range(L)):
            # Compute variables in cache need for backprop
            h_next = self.hiddens[t + 1][l, :, :]
            h_prev_t = self.hiddens[t][l, :, :]
            # h_prev_l can come from either input sequence or h_next from previous cell, l-1
            if l == 0:
                h_prev_l = self.x[:, t, :]
            else:
                h_prev_l = self.hiddens[t + 1][l - 1, :, :]
            # Done computing variables in cache needed for backprop
            # Perform backprop on RNN Cell
            h_prev_l, h_prev_t = self.rnn[l].backward(
                dh_next[l], h_next, h_prev_l, h_prev_t
            )
            # Update dh_next for the next previous time step
            dh_next[l] = h_prev_t
            # Add downstream gradient to the upstream gradient of the next previous cell step
            if l != 0:
                dh_next[l - 1] += h_prev_l
    return dh_next / N

if __name__ == "__main__":
    input_size = 3
    hidden_size = 4

```

```

output_size = 5
rnn = RNNPhonemeClassifier(input_size, hidden_size, output_size)

N = 10
seq_len = 3
x = np.random.random((N, seq_len, input_size))
logits = rnn.forward(x)
# print(logits.shape)
# print(logits)
delta = np.random.random((N, output_size))
dh = rnn.backward(delta)
print(dh)
print(dh.shape)

```

	shape
W_{xh}	(D, H)
W_{hh}	(H, H)
x_t	(N, D)
h	(N, H)
b_h	(H,)
z_t	(N, H)

3. RNN Cell (CS231n)

1. forward

1. $U = x_t \cdot W_{xh}$
2. $V = h_{t-1} \cdot W_{hh}$
3. $z_t = U + V + b_h$
4. $h_t = \tanh(z_t)$
5. Code: from rnn_layers.py

```

def rnn_step_forward(x, prev_h, Wx, Wh, b):
    """
    Run the forward pass for a single timestep of a vanilla RNN that uses a tanh
    activation function.

    The input data has dimension D, the hidden state has dimension H, and we use
    a minibatch size of N.

    Inputs:
    - x: Input data for this timestep, of shape (N, D).
    - prev_h: Hidden state from previous timestep, of shape (N, H)
    - Wx: Weight matrix for input-to-hidden connections, of shape (D, H)
    - Wh: Weight matrix for hidden-to-hidden connections, of shape (H, H)
    - b: Biases of shape (H,)

    Returns a tuple of:
    - next_h: Next hidden state, of shape (N, H)
    - cache: Tuple of values needed for the backward pass.
    """
    next_h, cache = None, None
    #####
    # TODO: Implement a single forward step for the vanilla RNN. Store the next #
    # hidden state and any values you need for the backward pass in the next_h #
    # and cache variables respectively. #
    #####
    z_t = x @ Wx + prev_h @ Wh + b
    next_h = np.tanh(z_t)
    # cache = (z_t, x, prev_h, Wx, Wh)
    cache = (next_h, x, prev_h, Wx, Wh)
    #####
    #                                     END OF YOUR CODE #
    #####
    return next_h, cache

```

2. backward

1. $\frac{\partial L}{\partial z_t} = \frac{\partial L}{\partial h_t} \odot (1 - \tanh^2(z_t))$
2. $\frac{\partial L}{\partial U} = \frac{\partial L}{\partial z_t}$
3. $\frac{\partial L}{\partial V} = \frac{\partial L}{\partial z_t}$
4. $\frac{\partial L}{\partial W_{xh}} = \left(\frac{\partial L}{\partial z_t} \cdot x_t \right)^T$
5. $\frac{\partial L}{\partial W_{hh}} = \left(\frac{\partial L}{\partial z_t} \cdot h_{t-1} \right)^T$
6. $\frac{\partial L}{\partial x_t} = \frac{\partial L}{\partial z_t} W_{xh}^T$
7. $\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial z_t} W_{hh}^T$

8. $\frac{\partial L}{\partial b} = (\frac{\partial L}{\partial z_t}).sum(axis=0)$

9. Code: from rnn_layers.py

```
def rnn_step_backward(dnext_h, cache):
    """
    Backward pass for a single timestep of a vanilla RNN.

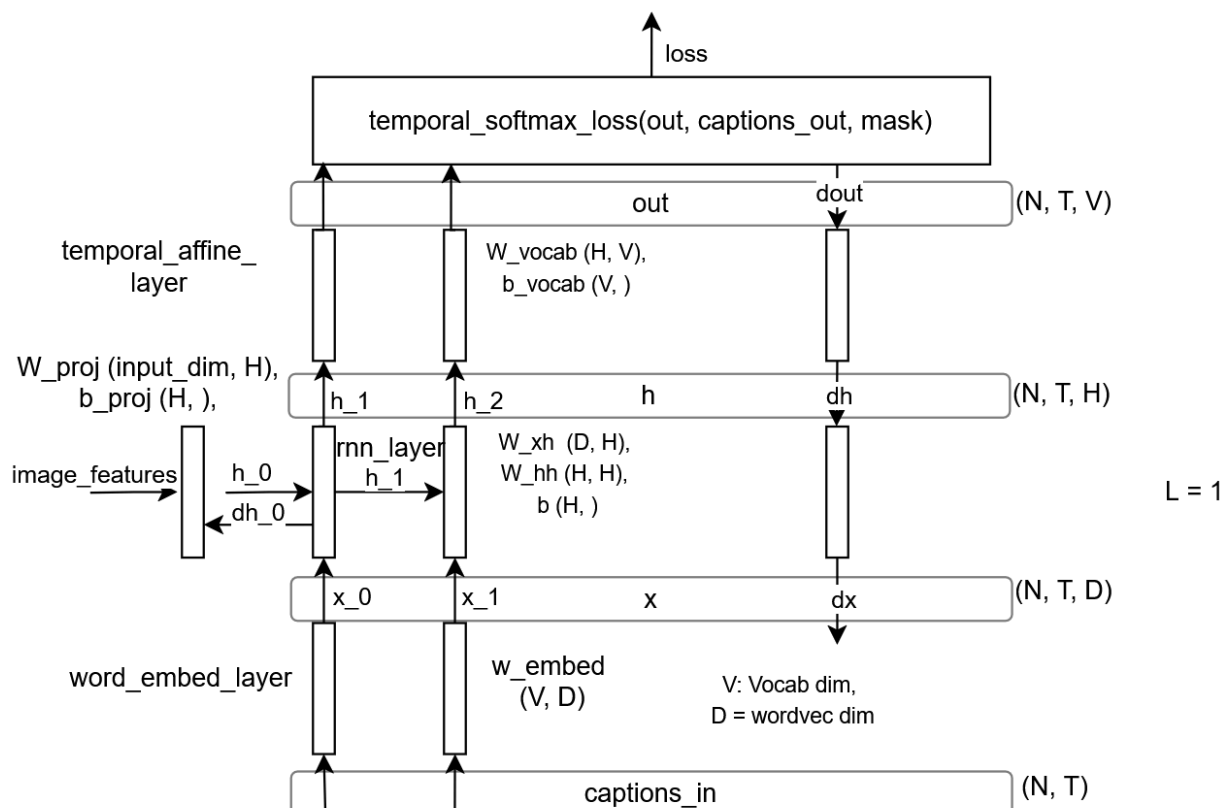
    Inputs:
    - dnext_h: Gradient of loss with respect to next hidden state
    - cache: Cache object from the forward pass

    Returns a tuple of:
    - dx: Gradients of input data, of shape (N, D)
    - dprev_h: Gradients of previous hidden state, of shape (N, H)
    - dWx: Gradients of input-to-hidden weights, of shape (D, H)
    - dWh: Gradients of hidden-to-hidden weights, of shape (H, H)
    - db: Gradients of bias vector, of shape (H,)
    """
    dx, dprev_h, dWx, dWh, db = None, None, None, None, None
    #####
    # TODO: Implement the backward pass for a single step of a vanilla RNN.
    #
    # HINT: For the tanh function, you can compute the local derivative in terms of
    # of the output value from tanh.
    #####
    # z_t, x_t, prev_h, Wx, Wh = cache
    next_h, x_t, prev_h, Wx, Wh = cache
    dz = dnext_h * (1 - next_h**2)
    dx = dz @ Wx.T
    dprev_h = dz @ Wh.T
    dWx = x_t.T @ dz
    dWh = prev_h.T @ dz
    db = np.sum(dz, axis=0)
    #####
    #                               END OF YOUR CODE                               #
    #####
    return dx, dprev_h, dWx, dWh, db
```

4. RNN Classifier (CS231n)

1. many-to-many architecture for image captioning:

simple RNN Classifier for Image Captioning



2. RNN Classifier forward,

3. RNN Classifier backward (BPTT),

4. Code: from rnn_layers.py

```
def rnn_forward(x, h0, Wx, Wh, b):
    """
    Run a vanilla RNN forward on an entire sequence of data. We assume an input
```


sequence composed of T vectors, each of dimension D. The RNN uses a hidden size of H, and we work over a minibatch containing N sequences. After running the RNN forward, we return the hidden states for all timesteps.

Inputs:

- x: Input data for the entire timeseries, of shape (N, T, D).
- h0: Initial hidden state, of shape (N, H)
- Wx: Weight matrix for input-to-hidden connections, of shape (D, H)
- Wh: Weight matrix for hidden-to-hidden connections, of shape (H, H)
- b: Biases of shape (H,)

Returns a tuple of:

- h: Hidden states for the entire timeseries, of shape (N, T, H).
- cache: Values needed in the backward pass

```
"""
h, cache = [], []
#####
# TODO: Implement forward pass for a vanilla RNN running on a sequence of #
# input data. You should use the rnn_step_forward function that you defined #
# above. You can use a for loop to help compute the forward pass.          #
#####
prev_h = h0
_, T, _ = x.shape
for t in range(T):
    next_h, next_cache = rnn_step_forward(
        x[:, t, :], prev_h,
        Wx, Wh, b
    )
    h.append(next_h)
    cache.append(next_cache)
    prev_h = next_h
h = np.array(h)
h = h.transpose(1, 0, 2)
#####
#                                     END OF YOUR CODE                       #
#####
return h, cache
```

```
def rnn_backward(dh, cache):
```

```
"""
```

Compute the backward pass for a vanilla RNN over an entire sequence of data.

Inputs:

- dh: Upstream gradients of all hidden states, of shape (N, T, H)

Returns a tuple of:

- dx: Gradient of inputs, of shape (N, T, D)
- dh0: Gradient of initial hidden state, of shape (N, H)
- dWx: Gradient of input-to-hidden weights, of shape (D, H)
- dWh: Gradient of hidden-to-hidden weights, of shape (H, H)
- db: Gradient of biases, of shape (H,)

```
"""
```

```
dx, dh0, dWx, dWh, db = None, None, None, None, None
```

```
#####
# TODO: Implement the backward pass for a vanilla RNN running an entire #
# sequence of data. You should use the rnn_step_backward function that you #
# defined above. You can use a for loop to help compute the backward pass. #
#####
N, T, H = dh.shape
```

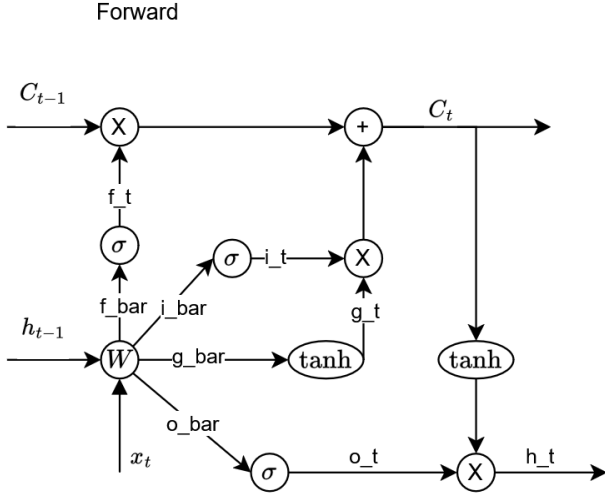
```
dx = []
# Extract D from x.shape
_, x, _, _ = cache[0]
_, D = x.shape
curr_dx = np.zeros((N, D))
dh0 = np.zeros((N, H))
dWx = np.zeros((D, H))
dWh = np.zeros((H, H))
db = np.zeros(H)
for t in reversed(range(T)):
    # Compute upstream gradients as input to RNN cell that computes local derivatives,
    # and produces dx, dprev_h, dWh, dWx and db
    if t == T - 1:
        dnext_h = dh[:, t, :]
    else:
        dnext_h = dh[:, t, :] + dprev_h
    curr_dx, dprev_h, prev_dWx, prev_dWh, prev_db = rnn_step_backward(dnext_h, cache[t])
    dWx += prev_dWx
    dWh += prev_dWh
    db += prev_db
    dx.append(curr_dx)
dh0 = dprev_h
# Because the way its elements are appended, resort the order based on ascending index of time
dx.reverse()
dx = np.array(dx)
dx = dx.transpose(1, 0, 2)
#####
#                                     END OF YOUR CODE                       #
#####
```

```
#####
return dx, dh0, dwx, dwh, db
```

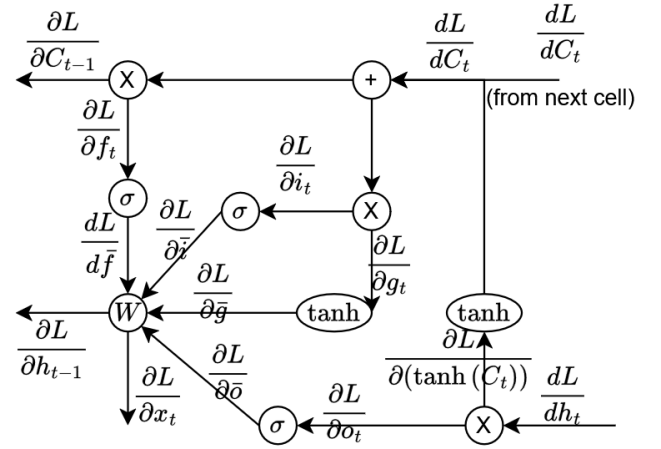
LSTM Cell

1. $\mathbf{i}_t = \sigma(\mathbf{W}_{ix}\mathbf{x}_t + \mathbf{W}_{ih}\mathbf{h}_{t-1})$,
2. $\mathbf{f}_t = \sigma(\mathbf{W}_{fx}\mathbf{x}_t + \mathbf{W}_{fh}\mathbf{h}_{t-1})$,
3. $\mathbf{o}_t = \sigma(\mathbf{W}_{ox}\mathbf{x}_t + \mathbf{W}_{oh}\mathbf{h}_{t-1})$,
4. $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{cx}\mathbf{x}_t + \mathbf{W}_{ch}\mathbf{h}_{t-1})$,
5. $\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$,

LSTM Cell



Backward



	shape
W_{xh}	(D, H)
W_{hh}	(H, H)
h_{t-1}	(N, H)
x_t	(N, D)
W	(D+H, 4H)
X	(N, H+D)
b	(4H,)
z	(N, 4H)

6. Forward

1. $i_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xi} \\ W_{hi} \end{bmatrix})$, input gate
2. $f_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xf} \\ W_{hf} \end{bmatrix})$, forget gate
3. $o_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xo} \\ W_{ho} \end{bmatrix})$, output gate
4. $g_t = \tanh([x_t, h_{t-1}] \begin{bmatrix} W_{xg} \\ W_{hg} \end{bmatrix})$, 'gate' gate

$$5. \begin{pmatrix} \bar{i} \\ \bar{f} \\ \bar{o} \\ \bar{g} \end{pmatrix} = [x_t \quad h_{t-1}] \begin{bmatrix} W_{xi} & W_{xf} & W_{xo} & W_{xg} \\ W_{hi} & W_{hf} & W_{ho} & W_{hg} \end{bmatrix} + b, \text{ shape: (n, 4h), where}$$

$$z = \begin{pmatrix} \bar{i} \\ \bar{f} \\ \bar{o} \\ \bar{g} \end{pmatrix}, \quad X = [x_t \quad h_{t-1}], \quad W = \begin{bmatrix} W_{xi} & W_{xf} & W_{xo} & W_{xg} \\ W_{hi} & W_{hf} & W_{ho} & W_{hg} \end{bmatrix}$$

$$6. i_t = \sigma(\bar{i})$$

$$7. f_t = \sigma(\bar{f})$$

8. $o_t = \sigma(\bar{o})$
9. $g_t = \tanh(\bar{g})$
10. $C_t = f_t \odot C_{t-1} + i_t \odot g_t$
11. $h_t = o_t \odot \tanh(C_t)$
12. Code: from rnn_layers.py

```
def lstm_step_forward(x, prev_h, prev_c, Wx, Wh, b):
    """
    Forward pass for a single timestep of an LSTM.

    The input data has dimension D, the hidden state has dimension H, and we use
    a minibatch size of N.

    Inputs:
    - x: Input data, of shape (N, D)
    - prev_h: Previous hidden state, of shape (N, H)
    - prev_c: previous cell state, of shape (N, H)
    - Wx: Input-to-hidden weights, of shape (D, 4H)
    - Wh: Hidden-to-hidden weights, of shape (H, 4H)
    - b: Biases, of shape (4H,)

    Returns a tuple of:
    - next_h: Next hidden state, of shape (N, H)
    - next_c: Next cell state, of shape (N, H)
    - cache: Tuple of values needed for backward pass.
    """
    next_h, next_c, cache = None, None, None
    #####
    # TODO: Implement the forward pass for a single timestep of an LSTM. #
    # You may want to use the numerically stable sigmoid implementation above. #
    #####
    x_stack = np.hstack((x, prev_h))
    w_stack = np.vstack((Wx, Wh))
    z = x_stack @ w_stack + b # shape: (N, 4*H)
    i_bar, f_bar, o_bar, g_bar = np.split(z, 4, axis=1)
    i_t = sigmoid(i_bar); f_t = sigmoid(f_bar); o_t = sigmoid(o_bar); g_t = np.tanh(g_bar)
    next_c = f_t * prev_c + i_t * g_t
    next_h = o_t * np.tanh(next_c)
    cache = (prev_h, x, prev_c, next_c,
             i_bar, f_bar, o_bar, g_bar,
             i_t, f_t, o_t, g_t,
             Wx, Wh)
    #####
    #                                     END OF YOUR CODE                                     #
    #####

    return next_h, next_c, cache
```

7. Backward

1. cache (to store in forward): $C_t, \bar{g}, \bar{o}, \bar{f}, \bar{i}, x_t, h_{t-1}, C_{t-1}, g_t, o_t, f_t, i_t, W_x, W_h$.
2. $A \cdot B$ denotes matrix multiplication, $A \odot B$ denotes element-wise multiplication, $\sigma(x) = \text{sigmoid}(x) = \frac{1}{1+e^{-x}}$.
3. $\frac{dL}{do_t} = \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial o_t} = \frac{\partial L}{\partial h_t} \odot \tanh(C_t)$
4. $\frac{\partial L}{\partial (\tanh C_t)} = \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial (\tanh C_t)} = \frac{\partial L}{\partial h_t} \odot o_t$
5. $\frac{\partial L}{\partial C_t} + = \frac{\partial L}{\partial (\tanh(C_t))} \frac{\partial (\tanh(C_t))}{\partial C_t} = \frac{\partial L}{\partial (\tanh C_t)} \odot (1 - \tanh^2 C_t)$
6. $\frac{\partial L}{\partial f_t} = \frac{\partial L}{\partial C_t} \odot C_{t-1}$
7. $\frac{\partial L}{\partial C_{t-1}} = \frac{\partial L}{\partial C_t} \odot f_t$
8. $\frac{\partial L}{\partial i_t} = \frac{\partial L}{\partial C_t} \odot g_t$
9. $\frac{\partial L}{\partial g_t} = \frac{\partial L}{\partial C_t} \odot i_t$
10. $\frac{\partial L}{\partial \bar{g}} = \frac{\partial L}{\partial g_t} \odot (1 - \tanh^2 \bar{g})$
11. $\frac{\partial L}{\partial \bar{o}} = \frac{\partial L}{\partial o_t} \frac{\partial \sigma(\bar{o})}{\partial \bar{o}} = \frac{\partial L}{\partial o_t} \odot \sigma(\bar{o}) \odot (1 - \sigma(\bar{o}))$
12. $\frac{\partial L}{\partial \bar{f}} = \frac{\partial L}{\partial f_t} \frac{\partial \sigma(\bar{f})}{\partial \bar{f}} = \frac{\partial L}{\partial f_t} \odot \sigma(\bar{f}) \odot (1 - \sigma(\bar{f}))$
13. $\frac{\partial L}{\partial \bar{i}} = \frac{\partial L}{\partial i_t} \frac{\partial \sigma(\bar{i})}{\partial \bar{i}} = \frac{\partial L}{\partial i_t} \odot \sigma(\bar{i}) \odot (1 - \sigma(\bar{i}))$
14. $\frac{\partial L}{\partial X} = \frac{\partial L}{\partial z} \cdot W^T$
15. $\frac{\partial L}{\partial W} = \left(\frac{\partial L}{\partial z}^T \cdot X \right)^T$
16. $\frac{\partial L}{\partial b} = \left(\frac{\partial L}{\partial z} \right) \cdot \text{sum}(\text{axis} = 0)$
17. Code: from rnn_layers.py

```
def lstm_step_backward(dnext_h, dnext_c, cache):
    """
    Backward pass for a single timestep of an LSTM.

    Inputs:
    - dnext_h: Gradients of next hidden state, of shape (N, H)
    - dnext_c: Gradients of next cell state, of shape (N, H)
```

```

- cache: Values from the forward pass

Returns a tuple of:
- dx: Gradient of input data, of shape (N, D)
- dprev_h: Gradient of previous hidden state, of shape (N, H)
- dprev_c: Gradient of previous cell state, of shape (N, H)
- dWx: Gradient of input-to-hidden weights, of shape (D, 4H)
- dWh: Gradient of hidden-to-hidden weights, of shape (H, 4H)
- db: Gradient of biases, of shape (4H,)
"""
dx, dh, dc, dWx, dWh, db = None, None, None, None, None, None
#####
# TODO: Implement the backward pass for a single timestep of an LSTM.      #
#                                                                            #
# HINT: For sigmoid and tanh you can compute local derivatives in terms of #
# the output value from the nonlinearity.                                  #
#####
prev_h, x, prev_c, next_c, i_bar, f_bar, o_bar, g_bar, i_t, f_t, o_t, g_t, Wx, Wh = cache
do_t = dnext_h * np.tanh(next_c)
dnext_c += dnext_h * o_t * (1 - np.tanh(next_c)**2)
df_t = dnext_c * prev_c
dprev_c = dnext_c * f_t
di_t = dnext_c * g_t
dg_t = dnext_c * i_t
dg_bar = dg_t * (1 - np.tanh(g_bar)**2)
do_bar = do_t * (1 - sigmoid(o_bar)) * sigmoid(o_bar)
df_bar = df_t * (1 - sigmoid(f_bar)) * sigmoid(f_bar)
di_bar = di_t * (1 - sigmoid(i_bar)) * sigmoid(i_bar)
dz = np.hstack((di_bar, df_bar, do_bar, dg_bar))
W = np.vstack((Wx, Wh))
dx_ = dz @ W.T
_, D = x.shape
_, H = prev_h.shape
dx = dx_[ :, :D]
dprev_h = dx_[ :, D:]
X = np.hstack((x, prev_h))
dW = (dz.T @ X).T
dWx = dW[ :D, :]
dWh = dW[D:, :]
db = dz.sum(axis=0)
#####
#                                END OF YOUR CODE                        #
#####

return dx, dprev_h, dprev_c, dWx, dWh, db

```

Flattening

1. Flattening dimensions is often used when we are doing matrix multiplications such as when preparing input to MLP.
2. During flattening the dimensions to be flattened are 'aggregated' into one dimension, for example (3, 4, 5) into (3, 20).
3. Code:

```

# torch.flatten

x = torch.rand(3, 4, 5)
a = x.flatten(1, 2)
b = x.flatten()
print("x.shape", x.shape)
print("a.shape", a.shape)
print("b.shape", b.shape)

```

4. Example:
 1. mfccs shape: (T, 28)
 2. during training using MLP:
 1. x: frames: (N, T, D)
 2. N: batch_size
 3. T: time_window: 2 * context_window + 1
 4. D: 28 (dimension)
 5. reshape x, (N, T * D)->mlp->phonemes (n, 42)

Incorporating information in neural nets

1. Most of the times in neural nets we need to incorporate information using either summing or concatenation.
2. Based on information theory, as long as the information is there it should be usable. From information standpoint it does not really matter either we use summing or concatenation.
 2. Sum: less memory requirement
 3. Concatenate: more memory

KL divergence

1. KL divergence: take the expectation of the logarithm of the ratio of p to q as if you were to assume that x is sampled through p .
2. $KL(P||Q) = \sum_{i=1}^n p_i \log(\frac{p_i}{q_i})$. p is the distribution we want to learn (target distribution), q is our model prediction distribution.

Cross entropy

1. In practice, $\text{delta} := \frac{\partial L}{\partial z}$ is computed inside the loss function e.g. CrossEntropyLoss class in classification tasks in pytorch. Also notice that delta is proportional to the error term $y_{pred} - y_{target}$. Because the Softmax function is already defined inside the CrossEntropyLoss, during model initialization the last layer of the model for classification is often a linear layer.

NLL

1. Negative Log-Likelihood error is equivalent to MSE if you have unit variance $\sigma^2 = I$.
2. If you learn the variance, then it ends up being weighted MSE.
3. Alternatively one can use LogSoftmax layer and apply NLLoss instead of CrossEntropyLoss.

Generative models

1. Given observed samples x from a distribution of interest, the goal of generative model is to learn to model its true distribution $p(x)$.
- 2.

Diffusion models

1. Connection to VAEs
 1. Diffusion models can be considered as special case of hierarchical VAEs.
 2. However, in diffusion models:
 1. The encoder is fixed.
 2. The latent variables have the same dimension as the data.
 3. The denoising model is shared across different timestep.
 4. The model is trained with some reweighting of the variational bound.
2. DDPM is an image-generation model
 1. Generating images.
 2. Sampling images from a simple prior.
 3. Modeling the distrib of the data X of interest.
 4. Computing the prob of data $p(x)$, x is an image.
 5. In contrast to discriminative models that models $p(y|x)$.
 6. Training: Diffusion to get the input.
 1. Sample an image x from the training set.
 2. Sample noise $\epsilon \sim \mathcal{N}(0, I)$.
 3. Sample a variance $\beta \in (0, 1)$.
 4. Get the input $x_t = \sqrt{1 - \beta}x + \sqrt{\beta}\epsilon$.
 5. reparametrization trick: $x = \text{mean} + \text{std_dev} * Z$ to sample from normal distribution. Z is drawn from a standard normal distribution.
 6. Note: reparametrization trick is a useful mental step to do backprop through a sampling process. While the gradient of an individual sample indeed does not exist or isn't meaningful, it does exist in expectation, and the expectation is all you need to backprop. Papers that refer this as a trick may come from the years 2012-2014.
 7. Training: Denoising to get the output
 1. Feed the input into the network.
 2. Network: U-Net f_θ .
 3. Predict the noise $f_\theta(x_t, t)$.
 4. Loss is MSE.
 5. input to model: $x_t = \sqrt{1 - \beta}x + \sqrt{\beta}\epsilon$
 6. output: $\tilde{\epsilon}, \hat{x} = \frac{x_t - \sqrt{\beta}\tilde{\epsilon}}{\sqrt{1 - \beta}}$, denoising: noisy image x_t minus the noise $\sqrt{\beta}\epsilon$.
 8. Inference: Iterative denoising
 9. DDPM does not learn the variance, just the mean of the noise ϵ .

References:

1. Unsupervised Representation Learning by Predicting Image Rotations. Gidaris, et. al. Ecole des Ponts ParisTech.
2. Understanding Diffusion Models: A Unified Perspective, Calvin Luo.
3. CS182 Deep Learning, UC Berkeley, Sergey Levine, 2021.
4. CS330 Variational Inference and Generative models, Chelsea Finn, Stanford, 2022.